

Data Fusion for Nowcasting

Roshni Sahoo
r.sahoo@columbia.edu

Columbia University

Abstract

Decision makers require real-time estimates of outcomes in a target population. We study real-time outcome estimation given representative but lagged data from target population and real-time but potentially biased data drawn from a survey population that differs from the target population. When the selection mechanism into the survey population is stationary, we demonstrate that the real-time mean outcome in the target population can be expressed as a time-invariant reweighting of the survey data. The weights are identified as the solution to an augmented convex optimization problem on data pooled from the target and survey populations across time.

1 Stationary Sampling Bias

Let P_t, Q_t denote the survey and target population distributions respectively at time-step t . Let Q_t be a distribution over an outcome $Y \in \mathcal{Y}$ and a binary response indicator $R \in \{0, 1\}$. Let P_t be the $Y \mid R = 1$ marginal of Q_t . We have access to P_t for $t = 1, \dots, M$ and Q_t for $t = 1, \dots, m$, where $m < M$. Define $\theta(t) := \mathbb{E}_{Q_t}[Y]$.

We propose to estimate $\theta(t)$ by leveraging the data from the paired survey and target distributions P_t, Q_t when $t = 1, 2, \dots, m$ and the survey distributions P_t when $t > m$. We define relative probability of sample selection between two individuals with outcome y and y' at time-step t as

$$w(y, y'; t) = \frac{\mathbb{P}_{Q_t}[R = 1 \mid Y = y]}{\mathbb{P}_{Q_t}[R = 1 \mid Y = y']}. \quad (1)$$

We consider learning in the presence of stationary sampling bias. Under stationary sampling bias, the relative probability of sample selection between individuals with different outcomes is constant over time.

Assumption 1 (Stationary Sampling Bias). The function $w(y, y'; t)$ is constant in t .

We highlight that under stationary sampling bias, data from multiple time steps can be pooled together to aid in estimating target parameters. As a preliminary building block, we define a pooled distribution F over our available data. Let T be a random variable that denotes the time step and let Z be a pseudolabel that captures whether a sample is drawn from the survey or target population. The distribution F is a distribution over (T, Z, Y) , where $Z \sim \text{Bernoulli}(1/2)$ and $T \sim \text{Uniform}([m])$, $F_{Y|T=t, Z=1} = Q_t$, and $F_{Y|T=t, Z=0} = P_t$.

We define a function class $\mathcal{H} = L^2(F_Y, \mathcal{Y})$, the space of square-integrable measurable functions with respect to F_Y and a function class $\mathcal{A} = L^2(F_T, \mathcal{T})$, the space of square-integrable measurable functions with respect to F_T .

Theorem 1. Suppose Assumption 1 holds. Define a loss function

$$L(r, a, z) := \log(1 + \exp(r + a)) - z \cdot (r + a). \quad (2)$$

Then, the target population mean at time-step $t \in [M]$ is given by

$$\theta(t) = \mathbb{E}_{P_t} [\exp(h^*(Y)) \cdot Y] / \mathbb{E}_{P_t} [\exp(h^*(Y))], \quad (3)$$

where h^* is the solution to the following augmented convex minimization problem

$$\{h^*(\cdot), \alpha^*(\cdot)\} \in \arg \min_{(h, \alpha) \in \mathcal{H} \times \mathcal{A}} \{\mathbb{E}_F [L(h(Y), \alpha(T), Z)] : \alpha(T) = 0\}. \quad (4)$$

[RS: why estimate the log density ratio instead of the density ratio? it's because then $r \cdot a$ is nonconvex hard to learn them]

Proof. Let $\gamma(y; t) = \mathbb{P}_{Q_t} [R = 1 \mid Y = y]$. First, we observe that Bayes rule implies that

$$dQ_t(y) = dP_t(y) \cdot \frac{1}{\gamma(y; t)} \cdot \mathbb{P}_{Q_t} [R = 1].$$

In addition, Bayes rule also implies that

$$\begin{aligned} \mathbb{E}_{P_t} \left[\frac{1}{\gamma(Y; t)} \right] &= \int_{\mathcal{Y}} \frac{\mathbb{P}_{Q_t} [Y = y \mid R = 1]}{\mathbb{P}_{Q_t} [R = 1 \mid Y = y]} dy \\ &= \int \frac{1}{\mathbb{P}_{Q_t} [R = 1]} \cdot dQ_t(y) \\ &= \frac{1}{\mathbb{P}_{Q_t} [R = 1]}. \end{aligned}$$

Thus, we have that

$$\theta(t) = \mathbb{E}_{P_t} \left[\frac{Y}{\gamma(Y; t)} \right] / \mathbb{E}_{P_t} \left[\frac{1}{\gamma(Y; t)} \right]. \quad (5)$$

As a result, it remains to show that right side of (5) is equal to right side (3). The main insight is that the solution to (4) is a function

$$\exp(h^*(y)) \propto \frac{1}{\gamma(y; t)} \quad (6)$$

up to a time-dependent scaling factor.

To characterize the solution to (4), we first demonstrate existence and uniqueness of the solution to a simpler unconstrained minimization problem. Let $\mathcal{S} = L^2(F_{Y,T}, \mathcal{Y} \times \mathcal{T})$, the space of square-integrable measurable functions with respect to $F_{Y,T}$. Set $\tilde{L}(s, z) = L(r, a, z)$ for any (r, a) such that $s = r + a$. Since L is strictly convex in $r + a$, $\tilde{L}$ is strictly convex in s . We consider the population risk minimization problem

$$\min_{s \in \mathcal{S}} \mathbb{E}_F [\tilde{L}(s(Y, T), Z)] \quad (7)$$

and note that it must have a unique minimizer s^* . The minimizer s^* must satisfy the KKT conditions conditional on $Y = y, T = t$ for every (y, t) . The KKT conditions are as follows

$$\nabla \mathbb{E}_F [L(s(Y, T), Z) \mid Y = y, T = t] = \frac{\exp(s(y, t))}{1 + \exp(s(y, t))} - \mathbb{P}_F [Z = 1 \mid Y = y, T = t] = 0.$$

Thus, $s^*(y, t)$ that solves the KKT conditions is given by

$$\begin{aligned} s^*(y, t) &= \log \left(\frac{\mathbb{P}_F [Z = 1 \mid Y = y, T = t]}{\mathbb{P}_F [Z = 0 \mid Y = y, T = t]} \right) \\ &= \log \left(\frac{dQ_t(y)}{dP_t(y)} \right). \end{aligned}$$

Now, we consider

$$\min_{(h, \alpha) \in \mathcal{H} \times \mathcal{A}} \mathbb{E}_F [L(h(Y), \alpha(T), Z)]. \quad (8)$$

We show that under Assumption 1, there exists a set of minimizers of the population risk $\mathbb{E}_F [L(h(Y), \alpha(T), Z)]$ over the set $\mathcal{H} \times \mathcal{A}$. We consider the set

$$\Theta^* = \{(h, \alpha) \in \mathcal{H} \times \mathcal{A} \mid h(y) + \alpha(t) = s^*(y, t)\}.$$

We verify that this set is nonempty under Assumption 1. Recall that $w(y, y'; t) = \frac{\gamma(y; t)}{\gamma(y'; t)}$ for all $y, y' \in \mathcal{Y}$ and $t \in \mathcal{T}$. Fix y' . Under Assumption 1, the left side does not depend on t and y' is fixed, so $w(y, y'; t) = \tilde{w}(y)$ and $\gamma(y'; t)$ is a constant that depends on t , so $\gamma(y'; t) = \lambda_t$. This implies $\gamma(y; t)$ is the product of a time-dependent constant and a function that only depends on y , i.e. $\gamma(y; t) = \lambda_t \cdot \tilde{w}(y)$. As a result, we have that $s^*(y, t)$ is given by

$$\begin{aligned} s^*(y, t) &= \log \left(\frac{dQ_t(y)}{dP_t(y)} \right) \\ &= \frac{\mathbb{P}_{Q_t} [R = 1]}{\gamma(y; t)} \\ &= \log \left(\frac{\mathbb{P}_{Q_t} [R = 1]}{\lambda_t \cdot \tilde{w}(y)} \right) \\ &= \alpha_t + h(y), \end{aligned}$$

where s^* is a time-dependent constant α_t and a function h that only depends on y . Thus, Θ^* must be nonempty. Furthermore, Θ^* is the set of minimizers of the unconstrained population risk because every $(h, \alpha) \in \Theta^*$ achieves the same population risk as s^* in (7).

We can show that there exists a unique $(h, \alpha) \in \Theta^*$ such that $\alpha(0) = 0$. To see this, suppose $(h_1, \alpha_1) \in \Theta^*$ and $\alpha_1(0) = c$, where $c \neq 0$. Then, we can define $h_2(y) = h_1(y) + c$ and $\alpha_2(t) = \alpha_1(t) - c$ so that $\alpha_2(0) = 0$. Thus, the constrained optimization problem in (4) has a solution that lies in the set of minimizers of the unconstrained optimization problem. Using an identical argument, we can also show that (4) has a unique solution.

Thus, the solution to (4) is equal to the log density ratio between the target and survey population at time t , i.e.

$$h^*(y) + \alpha^*(t) = \log \left(\frac{dQ_t(y)}{dP_t(y)} \right).$$

This implies that

$$\begin{aligned} \exp(h^*(y)) &= \exp(-\alpha^*(t)) \cdot \frac{dQ_t(y)}{dP_t(y)} \\ &= \exp(-\alpha^*(t)) \cdot \frac{\mathbb{P}_{Q_t} [R = 1]}{\gamma(y; t)} \\ &= C_t \cdot \frac{1}{\gamma(y; t)}. \end{aligned}$$

This yields (6), which proves the desired result. $\square$

Stationary sampling bias implies that there is a time-invariant reweighting function so that

$$\theta(t) = \frac{\mathbb{E}_{P_t} [g(Y) \cdot Y]}{\mathbb{E}_{P_t} [g(Y)]}.$$

This suggests that in the absence of individual-level samples from Q_t we can estimate g by solving a series of moment conditions. In particular, provided that we have access to lagged aggregate outcomes from the target population $\theta(t)$, we aim to find g that satisfies

$$\frac{\mathbb{E}_{P_t} [g(Y) \cdot Y]}{\mathbb{E}_{P_t} [g(Y)]} = \theta(t) \quad \forall t \in [m].$$

We can parametrize $g(y) = \exp(\gamma^T \psi(y))$ for $\gamma \in \mathbb{R}^m$. Then, we find γ so that

$$\frac{\mathbb{E}_{P_t} [\exp(\gamma^T \psi(Y)) \cdot Y]}{\mathbb{E}_{P_t} [\exp(\gamma^T \psi(Y))]} = \theta(t).$$

Equivalently, we solve

$$\min_{\gamma \in \mathbb{R}^m} \mathbf{1}^T \text{vec}(\log(\mathbb{E}_{P_t} [\exp(\gamma^T \psi(Y))])) - \theta(t)^T \gamma.$$

As a result, we have two ways to estimate the weights (1) using individual-level samples from Q_t and (2) using aggregate-level data from Q_t .

2 Nonstationary Sampling Bias

Under nonstationary sampling bias, the density ratio cannot be further decomposed into $\lambda(t) \cdot w(y)$.

$$\frac{dQ_t(y)}{dP_t(y)} = \lambda(t) \cdot w(y) \cdot r(y; t).$$

For simplicity, suppose that $\lambda(t) = 1$.

What does the pooling algorithm return when $\frac{dQ_t(y)}{dP_t(y)} = w(y) \cdot r(y; t)$?